CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. ATKT Examination (CL)

November 2013

CL211 Environmental Engineering-I

Date: 15.11.2013, Friday **Time: 01:30 p.m. To 04:30 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q-1 Select the right option for following questions **[07]**

1. Hardness in water is normally due to the presence of:
(A) trivalent metal cations (B) anions (C) divalent metal cations (D) none of these
2. MPN is a parameter that indicates _______ quality of water.
(A) chemical (B) biological (C) physical (D) social
3. Turbidity in water is measured by principle of light scattering. The unit of turbidity is:
(A) JTU (B) NTU (C) BTU (D) ATU
4. Stoke's law used for calculating settling velocity of a particle is valid only under:
(A) turbulent condition (B) laminar condition (C) transparent condition (D) none
5. Identify one of the following water quality parameters that is expressed as mg/L of CaCO3. (A) pH (B) acidity (c) COD (D) colour
6. Chlorine is used in water treatment as a:
(A) disinfectant (B) coagulant (C) flocculant (D) cleaning agent
7. Carbonate hardness is also known as _______ hardness.
(A) permanent (B) temporary (C) hard (D) pseudo

Q-2 (a) Enlist various methods used for population forecast and explain geometric increase method in detail. **[06]**

(b) The following is the population data of a city. Estimate the population of city in 2031 by geometric increase method. **[08]**

Year	1951	1961	1971	1981	1991	2001
Population (P)	138540	152680	172100	189800	197520	219820

OR

Q-2 (a) Define per capita water demand. Describe various factors affecting per capita water demand. **[07]**

(b) Why coagulation is used in water treatment? Explain various mechanisms of chemical coagulation **[07]**

Q-3 (a) Find out settling velocity of particles having specific gravity of 2.65 and diameter 0.04 mm at 30 °C. If a particle of diameter 10 mm has specific gravity of 0.85, will it ever settle down in water? **[05]**

(b) Explain in detail one of the following terms. **[04]**
1. Design Period, 2. Clariflocculator, 3. Rapid mix unit

(c) A primary settling tank is 20 m long and 5 m wide with side water depth of 3.0 m. If flow of water is 3000 m^3/d, calculate: 1) surface area and volume, 2) surface overflow rate, 3) detention period. **[05]**

OR

Q-3 (a) What are different types of settling? Explain each with neat figure. [08]

(b) Design a circular sedimentation tank to treat 2 million liters per day of coagulated water. Assume detention period to be 3h and water depth of 3.0 m. Calculate the surface overflow rate and weir loading rate for this tank. [06]

SECTION – II

Q-4 (a) Select the right option for following questions [07]

1. Which one of the following is not a disinfectant?
   (A) ozone (B) sodium hypochlorite (C) UV light (D) chloride
2. In backwashing operation of a sand filter, correct sequence of passing backwash water and compressed air through the sand bed is:
   (A) water followed by air (B) air followed by water (C) only air (D) only water
3. Disinfection is a process of killing
   (A) waterborne pathogens (B) all microorganisms (C) only viruses (D) only bacteria
4. Normally at the water inlet in an intake structure a _____ is provided to avoid large floating objects:
   (A) filter (B) bar screen (C) fabric filter (D) sedimentation basin
5. The trap which is provided to disconnect the house drain from the street sewer is:
   (A) P trap (B) interceptor trap (C) S-type trap (D) none of these
6. In a rectangular sedimentation tank (length L, width B, depth H) treating flow of Q, settling velocity of a particle is theoretically equal to:
   (A) $\frac{Q}{BxL}$ (B) $\frac{Q}{HxL}$ (C) $\frac{Q}{L}$ (D) $\frac{Q}{BxH}$
7. Settling velocity of a particle in water is affected by:
   (A) size and shape of particle (B) Sp. Gravity of solid (C) viscosity of water (D) all of these

Q-5 (a) Write short note (any one): (1) Natural ventilation system, (2) House service connection, (3) Man hole [06]

(b) Design a bell mouth canal intake for a city of 50000 persons drawing water from a canal which runs for 15 hours in a day with a depth of 1.8 m. Assume average water consumption = 140 LPCD. Assume velocity through screens and bell mouth to be less than 0.16 m/s and 0.32 m/s, respectively. [08]

OR

Q-5 (a) Differentiate between a rapid sand filter and a slow sand filter. [04]

(b) Briefly describe mechanisms of removal of suspended particles in a filter bed. [06]

(c) Write short note (any one): (1) Breakpoint chlorination (2) Types of water distribution system (3) Indoor air pollution [04]

Q-6 (a) It is required to supply water to a population of 30,000 at rate of 140 lpcd. The disinfectant used for chlorination is bleaching powder which contains 30% of available chlorine. Determine how much bleaching powder is required annually at the waterworks, if 0.3 ppm of chlorine dose is required for disinfection. If residual chlorine after 10 min contact time is 0.2 mg/L, what is chlorine demand? [06]

(b) Write short notes (any two): (1) Grid iron water distribution layout (2) Air conditioning, (3) Types of traps used in drainage, (4) Septic tank [08]

OR

Q-6 (a) Describe 'one pipe', 'two pipe' & 'single stack' systems of plumbing for buildings & state their merits & demerits. [07]

(b) Water is to be supplied to a town having population of 100000 at the rate of 140 LPCD from a river, 1 km away. The difference in the elevation between the lowest water level in the intake and service reservoir is 20 m. Determine the size of main. If the pumps are to be operated for total of 12h/day and the efficiency of pump is 80%, determine hp of pump required. Assume friction factor as 0.03 and velocity of flow through main = 1.5 m/s. Assume maximum demand = 1.8 times average demand of water. [07]

List of Useful formulae

$Pn = P(1+\frac{r}{100})^n$	$P = \mu G^2 V$	$SOR = \frac{Flow}{surface\ area}$	$v_s = \frac{gd^2}{18\nu}(Ss-1)$
$DT = \frac{V}{Q} = \frac{LBH}{Q}$	$Pipe\ flow, Q = \frac{\pi}{4}d^2 * v$	$v_s = 418(Ss-1)d^2(\frac{3t+70}{100})$	
$H_f = \frac{flv^2}{2gd}$	$HP = \frac{\rho QH}{75\eta}$		

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